

实验报告

实验项目名称 实验七 基本表的创建、插入、更新和删除
实 验 室 云计算研究中心实验室 520
所属课程名称 数 据 库
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【实验环境】

- Oracle11g 及以上

【实验目的】

- 学会基本表的创建;
- 掌握插入记录、更新记录和删除记录的操作。

【实验结果提交方式】

- 实验 7-3
按实验要求完成各个实验，并将各命令的执行过程以及执行成功后相应表的结构和数据屏幕复制下来，记录在本实验报告中。
- 将本实验报告存放在“XXXXXXXXXX-7.docx”文件中，其中“XXXXXXXXXX”是学号，并在教师规定的时间内通过 BB 系统提交该文件。

【实验 7-3 的实验结果】

记录各命令的执行过程以及执行成功后相应表的结构或数据。

(注意：自己创建下列题目涉及的表结构，插入的数据可参考 7-11 至 7-22)

前置任务，创建 Payment, Ptype, Orders, Order_items 表：

```
drop table Order_items;
```

```
drop table Product;
```

```
drop table Ptype;
```

```
drop table Orders;
```

```
drop table Payment;
```

```
drop table Customer;
```

```
CREATE TABLE Customer (
    CNO CHAR(5) PRIMARY KEY,
    CNAME NVARCHAR2(50),
    COMPANY NVARCHAR2(100),
    CITY NVARCHAR2(50),
    TEL VARCHAR(20)
);
```

```
CREATE TABLE Ptype (
    TNO CHAR(1) PRIMARY KEY,
    TNAME NVARCHAR2(15)
);
```

```
CREATE TABLE Product (  
    PNO char(4) PRIMARY KEY,  
    PNAME NVARCHAR2(100),  
    PRICE NUMBER(7, 2),  
    TNO CHAR(1),  
    INVENTORY INT,  
    FOREIGN KEY (TNO) REFERENCES Ptype(TNO)  
);  
  
CREATE TABLE Payment (  
    PAYMENT_TNO char(1) PRIMARY KEY,  
    PAYMENT_TYPE NVARCHAR2(50)  
);  
  
CREATE TABLE Orders (  
    ONO CHAR(5) PRIMARY KEY,  
    ORDER_DATE DATE,  
    CNO CHAR(5),  
    FREIGHT INT,  
    SHIPMENT_DATE DATE,  
    CITY NVARCHAR2(50),  
    PAYMENT_TNO char(1),  
    STATUS NVARCHAR2(20),  
    FOREIGN KEY (CNO) REFERENCES Customer(CNO),  
    FOREIGN KEY (PAYMENT_TNO) REFERENCES Payment(PAYMENT_TNO)  
);  
  
CREATE TABLE Order_items (  
    ONO CHAR(5),  
    PNO char(4),  
    QTY INT,  
    DISCOUNT DECIMAL(4, 2),  
    PRIMARY KEY (ONO, PNO),  
    FOREIGN KEY (ONO) REFERENCES Orders(ONO),  
    FOREIGN KEY (PNO) REFERENCES Product(PNO)
```

```
);
```

```
COMMIT;
```

1. 在 Payment 和 Ptype 表中插入数据。

```
INSERT INTO Ptype VALUES('1', 'Book');
```

```
INSERT INTO Ptype VALUES('2', 'CD');
```

```
INSERT INTO Ptype VALUES('3', 'Software');
```

```
INSERT INTO PAYMENT VALUES('1', 'Cash');
```

```
INSERT INTO PAYMENT VALUES('2', 'Check');
```

```
INSERT INTO PAYMENT VALUES('3', 'Credit Card');
```

```
INSERT INTO PAYMENT VALUES('4', 'Telegraphic Money');
```

```
SELECT * FROM Ptype;
```

	TNO	TNAME
1	1	Book
2	2	CD
3	3	Software

```
SELECT * FROM PAYMENT;
```

	PAYMENT_TNO	PAYMENT_TYPE
1	1	Cash
2	2	Check
3	3	Credit Card
4	4	Telegraphic Money

2. 利用 INSERT 语句在 Orders 表中直接插入 5 个记录。

```
ALTER SESSION SET NLS_DATE_FORMAT = 'YYYY-MM-DD HH24:MI:SS';
```

```
INSERT INTO ORDERS(ONO, ORDER_DATE, CNO, FREIGHT, SHIPMENT_DATE, CITY)
```

```
SELECT 'O0001', '2014-03-10', 'C0001', 8, '2014-03-11', 'Beijing' FROM DUAL UNION ALL
```

```
SELECT 'O0002', '2014-03-11', 'C0002', 8, '2014-03-12', 'Shanghai' FROM DUAL UNION ALL
```

```
SELECT 'O0003', '2014-03-11', 'C0009', 5, '2014-03-12', 'Shanghai' FROM DUAL UNION ALL
```

```
SELECT 'O0004', '2014-04-13', 'C0007', 5, '2014-04-15', 'Beijing' FROM DUAL UNION
```

ALL

```
SELECT 'O0005', '2014-04-14', 'C0010', 8, '2014-04-16', 'Beijing' FROM DUAL;
```

```
SELECT * FROM ORDERS;
```

	ONO	ORDER_DATE	CNO	FREIGHT	SHIPMENT_DATE	CITY	PAYMENT_TNO	STATUS
1	O0001	2014-03-10T00:00	C0001	8	2014-03-11T00:00	Beijing	(null)	(null)
2	O0002	2014-03-11T00:00	C0002	8	2014-03-12T00:00	Shanghai	(null)	(null)
3	O0003	2014-03-11T00:00	C0009	5	2014-03-12T00:00	Shanghai	(null)	(null)
4	O0004	2014-04-13T00:00	C0007	5	2014-04-15T00:00	Beijing	(null)	(null)
5	O0005	2014-04-14T00:00	C0010	8	2014-04-16T00:00	Beijing	(null)	(null)

3. 利用 INSERT 语句在 Orders 表中用交互方式插入 3 个记录。

```
INSERT INTO ORDERS(ONO, ORDER_DATE, CNO, FREIGHT, SHIPMENT_DATE, CITY,
PAYMENT_TNO, STATUS)
```

```
VALUES('&Ono', TO_DATE('&Order_Date', 'YYYY-MM-DD'), '&Cno', &Freight,
TO_DATE('&Shipment_Date', 'YYYY-MM-DD'), '&City', '&Payment_Tno', '&Status');
```

Name	Value
Ono	O0006
Order_Date	2014-04...
Cno	C0008
Freight	5
Shipment_Date	2014-04...
City	Shanghai
Payment_Tno	3
Status	Complete

	ONO	ORDER_DATE	CNO	FREIGHT	SHIPMENT_DATE	CITY	PAYMENT_TNO	STATUS
1	O0007	2014-05-26T00:00	C0010	8	2014-05-28T00:00	Shanghai	3	Complete
2	O0008	2014-06-17T00:00	C0006	5	2014-06-18T00:00	Beijing	1	Complete
3	O0001	2014-03-10T00:00	C0001	8	2014-03-11T00:00	Beijing	(null)	(null)
4	O0002	2014-03-11T00:00	C0002	8	2014-03-12T00:00	Shanghai	(null)	(null)
5	O0003	2014-03-11T00:00	C0009	5	2014-03-12T00:00	Shanghai	(null)	(null)
6	O0004	2014-04-13T00:00	C0007	5	2014-04-15T00:00	Beijing	(null)	(null)
7	O0005	2014-04-14T00:00	C0010	8	2014-04-16T00:00	Beijing	(null)	(null)
8	O0006	2014-04-25T00:00	C0008	5	2014-04-26T00:00	Shanghai	3	Complete

4. 设置 Orders 表前 5 行数据中的 PAYMENT_TNO 和 STATUS 属性的值。

```
UPDATE ORDERS
```

```
SET PAYMENT_TNO = '1', STATUS = 'Complete'
```

```
WHERE ONO IN ('O0001', 'O0004', 'O0005');
```

```
UPDATE ORDERS
```

```
SET PAYMENT_TNO = '2', STATUS = 'Complete'
```

```
WHERE ONO IN ('O0002', 'O0003');
```

```
SELECT * from ORDERS;
```

	ONO	ORDER_DATE	CNO	FREIGHT	SHIPMENT_DATE	CITY	PAYMENT_TNO	STATUS
1	00007	2014-05-26T00:00	C0010	8	2014-05-28T00:00	Shanghai	3	Complete
2	00008	2014-06-17T00:00	C0006	5	2014-06-18T00:00	Beijing	1	Complete
3	00001	2014-03-10T00:00	C0001	8	2014-03-11T00:00	Beijing	1	Complete
4	00002	2014-03-11T00:00	C0002	8	2014-03-12T00:00	Shanghai	2	Complete
5	00003	2014-03-11T00:00	C0009	5	2014-03-12T00:00	Shanghai	2	Complete
6	00004	2014-04-13T00:00	C0007	5	2014-04-15T00:00	Beijing	1	Complete
7	00005	2014-04-14T00:00	C0010	8	2014-04-16T00:00	Beijing	1	Complete
8	00006	2014-04-25T00:00	C0008	5	2014-04-26T00:00	Shanghai	3	Complete

5. 在 Product 和 Order_items 表中插入数据。

先插入两行 Orders 数据:

```
INSERT INTO ORDERS(ONO, ORDER_DATE, CNO, FREIGHT, SHIPMENT_DATE, CITY,
PAYMENT_TNO, STATUS)
```

```
VALUES ('O0009', TO_DATE('2014-07-21', 'YYYY-MM-DD'), 'C0008', 5,
TO_DATE('2014-07-22', 'YYYY-MM-DD'), 'Shanghai', '2','in process');
```

```
INSERT INTO ORDERS (ONO, ORDER_DATE, CNO, FREIGHT, SHIPMENT_DATE, CITY,
PAYMENT_TNO, STATUS)
```

```
VALUES ('O0010', TO_DATE('2014-07-23', 'YYYY-MM-DD'), 'C0005', 5,
TO_DATE('2014-07-25', 'YYYY-MM-DD'), 'Beijing', '1','in process');
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('1001',
'Advanced Marketing', 20.5, '1', 120);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('1002', 'Visual
Basic Programming', 28, '1', 200);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('1003',
'Computer Application', 30.55, '1', 80);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('1004', 'An
Introduction to Database Systems', 20, '1', 12);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('1005',
'Microeconomics', 35.8, '1', 150);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('2001', 'The
Lion King', 35, '2', 150);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('2002', 'Classic
Disney', 25, '2', 20);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('3001',
'Microsoft Money 2010', 70.5, '3', 300);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('3002',
'Microsoft Student 2009', 80, '3', 150);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('3003',
'Norton AntiVirus 2014', 40.9, '3', 250);
```

```
INSERT INTO Product (PNO, PNAME, PRICE, TNO, INVENTORY) VALUES ('3004', 'Math
Advantage 2007', 30, '3', 10);
```

	PNO	PNAME	PRICE	TNO	INVENTORY
1	1001	Advanced Marketing	20.5	1	120
2	1002	Visual Basic Programming	28	1	200
3	1003	Computer Application	30.55	1	80
4	1004	An Introduction to Database Systems	20	1	12
5	1005	Microeconomics	35.8	1	150
6	2001	The Lion King	35	2	150
7	2002	Classic Disney	25	2	20
8	3001	Microsoft Money 2010	70.5	3	300

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0001', '1001', 5,
NULL);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0001', '1002', 1,
0.2);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0001', '1003', 3,
0.3);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0001', '2001', 1,
0.2);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0001', '2002', 1,
0.2);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0002', '1001', 2,
0.0);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0002', '1004', 5,
0.4);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0002', '1005', 1,
0.05);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0002', '3003', 3,
0.3);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0003', '1005', 1,
0.05);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0004', '1005', 1,
0.05);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0005', '1005', 1,
0.05);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0006', '1004', 5,
0.4);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0006', '1005', 1,
null);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0006', '2001', 2,
0.3);
```

```
INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0006', '2002', 1,
```

```

0.2);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0006', '3003', 2,
0.3);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0007', '1004', 1,
0.4);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0007', '3004', 3,
0.2);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0008', '1002', 1,
0.2);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0008', '2001', 3,
0.3);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0009', '3003', 2,
0.3);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0009', '1001', 1,
null);

INSERT INTO ORDER_ITEMS (ONO, PNO, QTY, DISCOUNT) VALUES ('O0010', '1005', 1,
0.05);

```

	ONO	PNO	QTY	DISCOUNT
1	00001	1001	5	(null)
2	00001	1002	1	0.2
3	00001	1003	3	0.3
4	00001	2001	1	0.2
5	00001	2002	1	0.2
6	00002	1001	2	0
7	00002	1004	5	0.4
8	00002	1005	1	0.05
9	00002	3003	3	0.3
10	00003	1005	1	0.05
11	00004	1005	1	0.05
12	00005	1005	1	0.05
13	00006	1004	5	0.4
14	00006	1005	1	(null)
15	00006	2001	2	0.3
16	00006	2002	1	0.2
17	00006	3003	2	0.3
18	00007	1004	1	0.4
19	00007	3004	3	0.2
20	00008	1002	1	0.2
21	00008	2001	3	0.3
22	00009	3003	2	0.3
23	00009	1001	1	(null)
24	00010	1005	1	0.05

```
COMMIT;
```


6. 参阅错误!未找到引用源。给出的数据，修改 Product 表的产品库存。

```
UPDATE PRODUCT
SET INVENTORY = 110
WHERE PNO = '1001';
```

```
UPDATE PRODUCT
SET INVENTORY = 190
WHERE PNO = '1002';
```

```
UPDATE PRODUCT
SET INVENTORY = 70
WHERE PNO = '1003';
```

```
UPDATE PRODUCT
SET INVENTORY = 2
WHERE PNO = '1004';
```

```
UPDATE PRODUCT
SET INVENTORY = 140
WHERE PNO = '1005';
```

	PNO	PNAME	PRICE	TNO	INVENTORY
1	1001	Advanced Marketing	20.5	1	110
2	1002	Visual Basic Programming	28	1	190
3	1003	Computer Application	30.55	1	70
4	1004	An Introduction to Database Systems	20	1	2
5	1005	Microeconomics	35.8	1	140
6	2001	The Lion King	35	2	150
7	2002	Classic Disney	25	2	20
8	3001	Microsoft Money 2010	70.5	3	300
9	3002	Microsoft Student 2009	80	3	150
10	3003	Norton AntiVirus 2014	40.9	3	250
11	3004	Math Advantage 2007	30	3	10

7. 删除与 “1001” 号产品有关的所有信息。

```
DELETE
FROM Order_items
where PNO = '1001';
```

	ONO	PNO	QTY	DISCOUNT
1	00001	1002	1	0.2
2	00001	1003	3	0.3
3	00001	2001	1	0.2
4	00001	2002	1	0.2
5	00002	1004	5	0.4
6	00002	1005	1	0.05
7	00002	3003	3	0.3
8	00003	1005	1	0.05
9	00004	1005	1	0.05
10	00005	1005	1	0.05
11	00006	1004	5	0.4
12	00006	1005	1	(null)
13	00006	2001	2	0.3
14	00006	2002	1	0.2
15	00006	3003	2	0.3
16	00007	1004	1	0.4
17	00007	3004	3	0.2
18	00008	1002	1	0.2
19	00008	2001	3	0.3
20	00009	3003	2	0.3
21	00010	1005	1	0.05

DELETE

from Product

where PNO = '1001';

	PNO	PNAME	PRICE	TNO	INVENTORY
1	1002	Visual Basic Programming	28	1	190
2	1003	Computer Application	30.55	1	70
3	1004	An Introduction to Database Systems	20	1	2
4	1005	Microeconomics	35.8	1	140
5	2001	The Lion King	35	2	150
6	2002	Classic Disney	25	2	20
7	3001	Microsoft Money 2010	70.5	3	300
8	3002	Microsoft Student 2009	80	3	150
9	3003	Norton AntiVirus 2014	40.9	3	250
10	3004	Math Advantage 2007	30	3	10

8. 使用“CREATE TABLE ... AS”语句从 Product 表创建另一个名称为 Product1 的表，其中包含所有库存小于 100 的产品信息。

CREATE TABLE Product1 AS

SELECT * FROM Product WHERE INVENTORY < 100;

SELECT * FROM Product1;

	PNO	PNAME	PRICE	TNO	INVENTORY
1	1003	Computer Application	30.55	1	70
2	1004	An Introduction to Database Systems	20	1	2
3	2002	Classic Disney	25	2	20
4	3004	Math Advantage 2007	30	3	10

9. 将 Product1 表中库存小于 50 的产品的库存增加 200。

UPDATE PRODUCT1

```
SET INVENTORY = INVENTORY + 200
```

```
WHERE INVENTORY < 50;
```

```
SELECT * FROM PRODUCT1;
```

```
COMMIT;
```

	PNO	PNAME	PRICE	TNO	INVENTORY
1	1003	Computer Application	30.55	1	70
2	1004	An Introduction to Database Systems	20	1	202
3	2002	Classic Disney	25	2	220
4	3004	Math Advantage 2007	30	3	210

10. 将 Employee 表中所有办事员（CLERK）的薪金提高 5%。

```
UPDATE Employee
```

```
SET sal = sal * 1.05
```

```
where JOB = 'CLERK';
```

```
COMMIT;
```

```
select * from EMPLOYEE where job = 'CLERK';
```

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7369	SMITH	CLERK	7902	1980-12-17T00:00	840	(null)	20
2	7876	ADAMS	CLERK	7788	1987-07-13T00:00	1155	(null)	20
3	7900	JAMES	CLERK	7698	1981-12-03T00:00	997.5	(null)	30
4	7934	MILLER	CLERK	7782	1982-01-23T00:00	1365	(null)	10

11. 将 Employee 表中部门“30”的销售员（SALESMAN）的薪金增加 300 元。

```
UPDATE EMPLOYEE
```

```
set sal = sal + 300
```

```
where deptno = 30 and job = 'SALESMAN';
```

```
COMMIT;
```

```
SELECT * FROM EMPLOYEE WHERE DEPTNO = 30 AND JOB = 'SALESMAN';
```

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7499	ALLEN	SALESMAN	7698	1981-02-20T00:00	1900	300	30
2	7521	WARD	SALESMAN	7698	1981-02-22T00:00	1550	500	30
3	7654	MARTIN	SALESMAN	7698	1981-09-28T00:00	1550	1400	30
4	7844	TURNER	SALESMAN	7698	1981-09-08T00:00	1800	0	30

12. 在 Employee 表中将“7369”号雇员从部门“20”转到部门“30”。

```
UPDATE EMPLOYEE
```

```
SET DEPTNO = 30
```

WHERE EMPNO = 7369;

SELECT * FROM EMPLOYEE where EMPNO=7369;

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7369	SMITH	CLERK	7902	1980-12-17T00:00	840	(null)	30

13. 在 Employee 表中，将 2000 年以前雇佣的职员的薪金提高 1000 元。

UPDATE EMPLOYEE

SET SAL = SAL + 1000

WHERE EXTRACT(YEAR FROM HIREDATE) < 2000;

select * from EMPLOYEE;

COMMIT;

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7369	SMITH	CLERK	7902	17/12/1980 12:00:00	1840	(null)	20
2	7499	ALLEN	SALESMAN	7698	20/02/1981 12:00:00	2900	300	30
3	7521	WARD	SALESMAN	7698	22/02/1981 12:00:00	2550	500	30
4	7566	JONES	MANAGER	7839	02/04/1981 12:00:00	3975	(null)	20
5	7654	MARTIN	SALESMAN	7698	28/09/1981 12:00:00	2550	1400	30
6	7698	BLAKE	MANAGER	7839	01/05/1981 12:00:00	3850	(null)	30
7	7782	CLARK	MANAGER	7839	09/06/1981 12:00:00	3450	(null)	10
8	7788	SCOTT	ANALYST	7566	13/07/1987 12:00:00	4000	(null)	20
9	7839	KING	PRESIDENT	(null)	17/11/1981 12:00:00	6000	(null)	10
10	7844	TURNER	SALESMAN	7698	08/09/1981 12:00:00	2800	0	30
11	7876	ADAMS	CLERK	7788	13/07/1987 12:00:00	2155	(null)	20
12	7900	JAMES	CLERK	7698	03/12/1981 12:00:00	1997.5	(null)	30
13	7902	FORD	ANALYST	7566	03/12/1981 12:00:00	4000	(null)	20
14	7934	MILLER	CLERK	7782	23/01/1982 12:00:00	2365	(null)	10

14. 在 Employee 表中删除佣金为 0 的销售员（SALESMAN）的信息。

DELETE FROM EMPLOYEE

where job = 'SALESMAN' and comm = 0;

SELECT * FROM EMPLOYEE where job = 'SALESMAN';

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	7499	ALLEN	SALESMAN	7698	20/02/1981 12:00:00	2900	300	30
2	7521	WARD	SALESMAN	7698	22/02/1981 12:00:00	2550	500	30
3	7654	MARTIN	SALESMAN	7698	28/09/1981 12:00:00	2550	1400	30

15. 删除部门“10”的部门信息，以及其中雇员的信息。

DELETE FROM EMPLOYEE WHERE DEPTNO = 10;

DELETE FROM DEPARTMENT WHERE DEPTNO = 10;

SELECT * FROM EMPLOYEE;

SELECT * FROM DEPARTMENT;

	DEPTNO	DNAME	LOC
1	20	RESEARCH	DALLAS
2	30	SALES	CHICAGO
3	40	OPERATIONS	BOSTON

- COMMIT;

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
-------	-------	-----	-----	----------	-----	------	--------

- ```
SELECT TABLE_NAME, NUM_ROWS, TABLESPACE_NAME FROM USER_TABLES;
SELECT COLUMN_NAME, DATA_TYPE, DATA_LENGTH FROM USER_TAB_COLUMNS;
```

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|    | COLUMN_NAME   | DATA_TYPE | DATA_LENGTH |
|----|---------------|-----------|-------------|
| 1  | CNO           | CHAR      | 5           |
| 2  | CNAME         | NVARCHAR2 | 100         |
| 3  | COMPANY       | NVARCHAR2 | 200         |
| 4  | CITY          | NVARCHAR2 | 100         |
| 5  | TEL           | VARCHAR2  | 20          |
| 6  | DEPTNO        | NUMBER    | 22          |
| 7  | DNAME         | VARCHAR2  | 14          |
| 8  | LOC           | VARCHAR2  | 13          |
| 9  | EMPNO         | NUMBER    | 22          |
| 10 | ENAME         | VARCHAR2  | 10          |
| 11 | JOB           | VARCHAR2  | 9           |
| 12 | MGR           | NUMBER    | 22          |
| 13 | HIREDATE      | DATE      | 7           |
| 14 | SAL           | NUMBER    | 22          |
| 15 | COMM          | NUMBER    | 22          |
| 16 | DEPTNO        | NUMBER    | 22          |
| 17 | ONO           | CHAR      | 5           |
| 18 | ORDER_DATE    | DATE      | 7           |
| 19 | CNO           | CHAR      | 5           |
| 20 | FREIGHT       | NUMBER    | 22          |
| 21 | SHIPMENT_DATE | DATE      | 7           |
| 22 | CITY          | NVARCHAR2 | 100         |
| 23 | PAYMENT_TNO   | CHAR      | 1           |
| 24 | STATUS        | NVARCHAR2 | 40          |
| 25 | ONO           | CHAR      | 5           |
| 26 | PNO           | CHAR      | 4           |
| 27 | QTY           | NUMBER    | 22          |
| 28 | DISCOUNT      | NUMBER    | 22          |
| 29 | PAYMENT_TNO   | CHAR      | 1           |
| 30 | PAYMENT_TYPE  | NVARCHAR2 | 100         |
| 31 | PNO           | CHAR      | 4           |
| 32 | PNAME         | NVARCHAR2 | 200         |
| 33 | PRICE         | NUMBER    | 22          |
| 34 | TNO           | CHAR      | 1           |
| 35 | INVENTORY     | NUMBER    | 22          |
| 36 | PNO           | CHAR      | 4           |
| 37 | PNAME         | NVARCHAR2 | 200         |
| 38 | PRICE         | NUMBER    | 22          |
| 39 | TNO           | CHAR      | 1           |
| 40 | INVENTORY     | NUMBER    | 22          |
| 41 | TNO           | CHAR      | 1           |
| 42 | TNAME         | NVARCHAR2 | 30          |

**【实验思考】**

1. 如何用 NLS\_DATE\_FORMAT 参数设置默认的日期格式？
2. 如何一次在一个基本表中插入多行数据？
3. 举例说明如何进行表复制操作？

**【思考结果】**

将思考结果记录在本实验报告中。

1. 如何用 NLS\_DATE\_FORMAT 参数设置默认日期格式？

```
ALTER SESSION SET NLS_DATE_FORMAT = 'YYYY-MM-DD HH24:MI:SS';
```

也可以再获取管理员权限后做系统级别的修改

还可以在 Windows 系统环境变量中修改

2. 如何一次在一个基本表中插入多行数据？

使用 INSERT ALL 语句

这是 Oracle 特有的语法，适用于大多数版本。注意最后必须包含 SELECT \* FROM dual。

```
INSERT ALL
 INTO my_table (id, name) VALUES (1, 'Alice')
 INTO my_table (id, name) VALUES (2, 'Bob')
 INTO my_table (id, name) VALUES (3, 'Charlie')
SELECT * FROM dual;
```

也可以使用 UNION ALL

例如在 7-3-2 中：

```
INSERT INTO ORDERS(ONO, ORDER_DATE, CNO, FREIGHT, SHIPMENT_DATE, CITY)
SELECT '00001', TO_DATE('2014-03-10', 'YYYY-MM-DD'), 'C0001', 8,
TO_DATE('2014-03-11', 'YYYY-MM-DD'), 'Beijing' FROM DUAL UNION ALL
SELECT '00002', TO_DATE('2014-03-11', 'YYYY-MM-DD'), 'C0002', 8,
TO_DATE('2014-03-12', 'YYYY-MM-DD'), 'Shanghai' FROM DUAL UNION ALL
SELECT '00003', TO_DATE('2014-03-11', 'YYYY-MM-DD'), 'C0009', 5,
TO_DATE('2014-03-12', 'YYYY-MM-DD'), 'Shanghai' FROM DUAL UNION ALL
SELECT '00004', TO_DATE('2014-04-13', 'YYYY-MM-DD'), 'C0007', 5,
TO_DATE('2014-04-15', 'YYYY-MM-DD'), 'Beijing' FROM DUAL UNION ALL
SELECT '00005', TO_DATE('2014-04-14', 'YYYY-MM-DD'), 'C0010', 8,
TO_DATE('2014-04-16', 'YYYY-MM-DD'), 'Beijing' FROM DUAL;
```

```
SELECT * from ORDERS;
```

```
Commit;
```

3. 举例说明如何进行表复制操作？

使用 CREATE TABLE AS SELECT 语句。

比如 7-3-8 中：

```
CREATE TABLE Product1 AS
 SELECT * FROM Product WHERE INVENTORY < 100;
```

实验成绩：\_\_\_\_\_ 批阅老师：\_\_\_\_\_ 批阅日期：\_\_\_\_\_